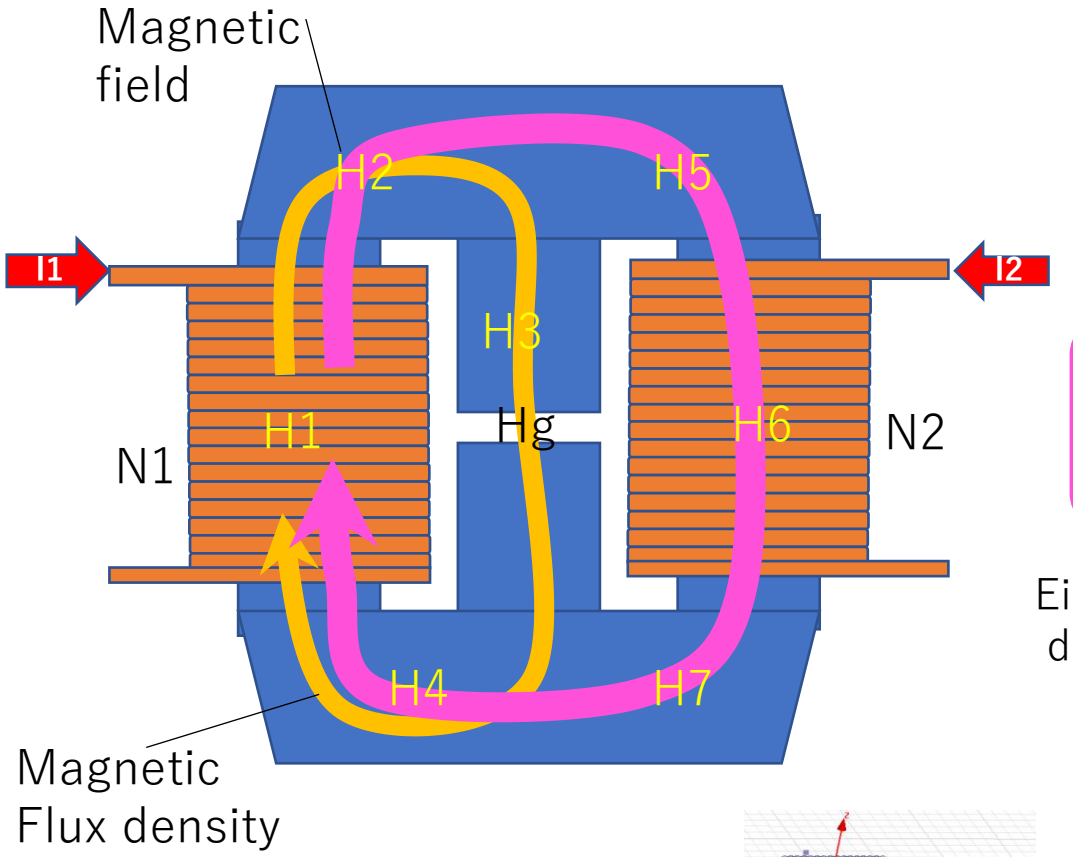
Algebraic Solution Method for Nonlinear Coupled Magnetic Circuits

Kazuhisa ishizuka

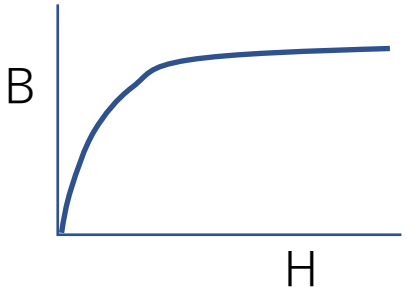
Reactor State at a Given Instant



Ampere's Law I1 x N1=H1 ℓ 1+H2 ℓ 2+H3 ℓ 3+Hg ℓ g+H4 ℓ 4
Flux Continuity Law B1xA1 = B2xA2 = B4xA4

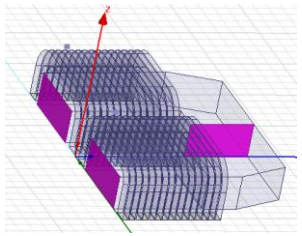
Ampere's Law I1xN1-I2xN2=H1 ℓ 1+H2 ℓ 2+H5 ℓ 5+H7 ℓ 7+H4 ℓ 4
Flux Continuity Law B5xA5 = B6xA6 = B7xA7

Eight equations for eight unknowns (B1 to Bg), resulting in a uniquely determined magnetic state.



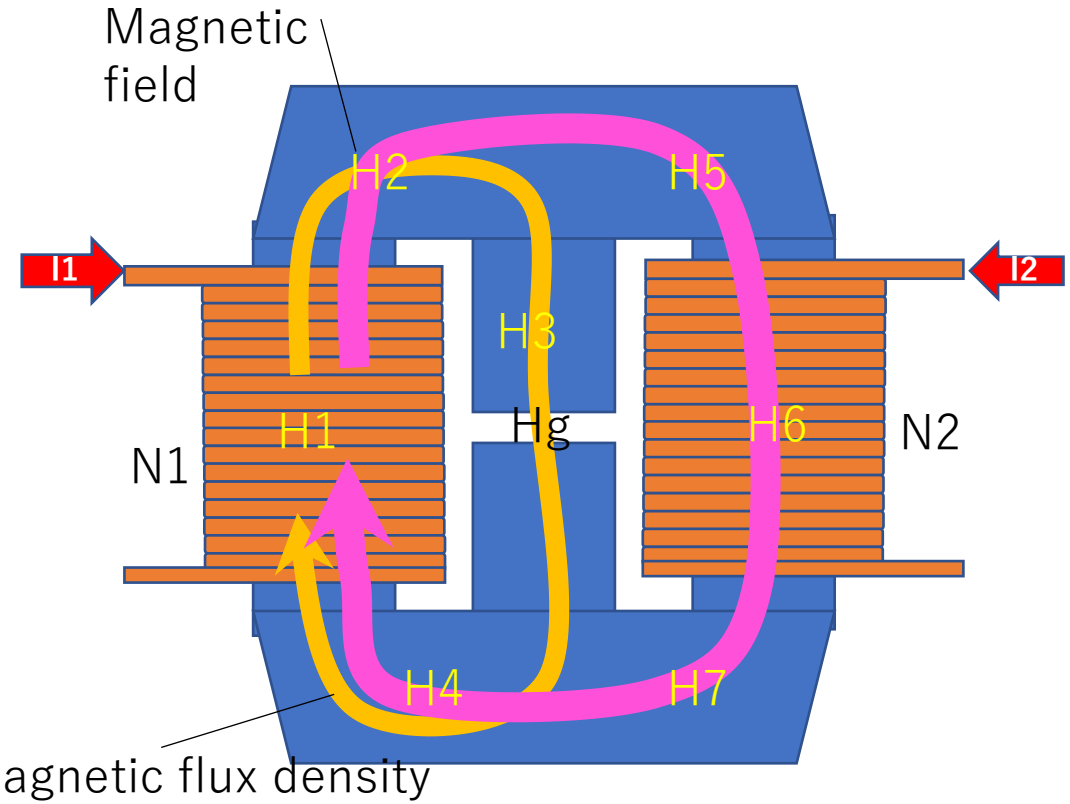
However, B and H are not linearly proportional (material-specific B-H characteristic).

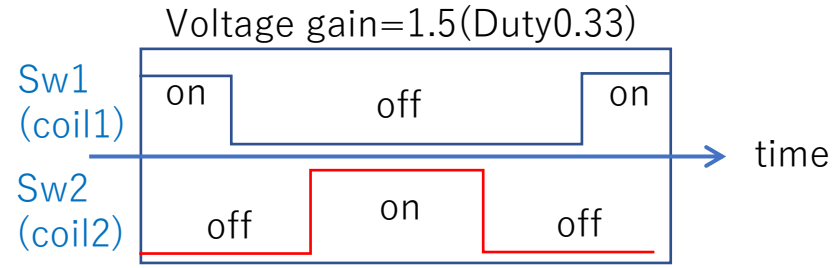
->**Solve the resulting nonlinear system of equations numerically**

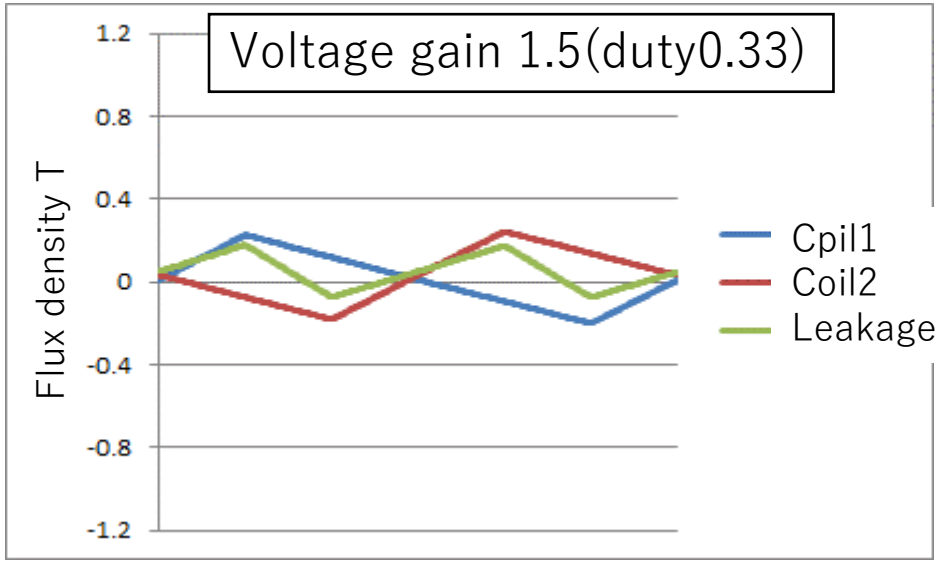


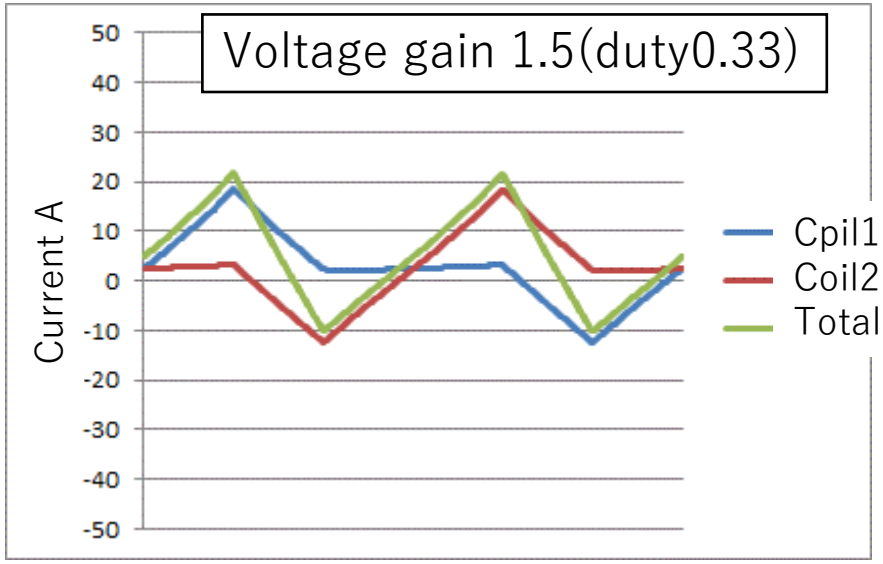
Flux linkage of Coil 1: λ 1 = N1xB1xA1

N1,N2 Number of coil turns
I1,I2 Current at a given instant
B1,B2…B7,Bg Magnetic flux density at a given instant
A1,A2～A7,Ag Cross-sectional area of each magnetic path segment
ℓ 1, ℓ 2… ℓ 7, ℓ g Magnetic path length of each segment









The voltage across each coil is determined by the circuit operating state.

Since the applied voltage uniquely determines the rate of change of flux, the current at each time step can be calculated sequentially.